Chapter

7 Waves

What is a tsunami? A tsunami, or seismic sea wave, is a series of water waves started by an earthquake or volcano. They differ from the usual waves found in oceans with much longer wavelengths, up to 200 km, and much greater speeds, over 800 km h^{-1}. The amplitude of these waves increase in shallower water and the vast quantity of energy that they transfer is dissipated when they meet solid objects. There are now deep-ocean and coastal automatic warning systems in many parts of the world affected by tsunamis.

Prior understanding

From previous courses you may remember some basic information about how to describe waves, such as wavelength, amplitude or frequency. You may also recall some examples of waves, such as sound and those that make up the electromagnetic spectrum. In addition, you may remember some properties of waves such as their ability to be reflected and refracted.

Learning aims

In this chapter you will learn what is meant by wave motion and how to describe waves in detail. You will discover the differences between longitudinal and transverse waves, and see how longitudinal sound waves can be represented on an oscilloscope. You will learn about the intensity of progressive waves. The Doppler effect will be introduced and you will see how it affects different observers' perception of sound waves. You will also learn more about the electromagnetic spectrum, and develop an understanding of polarisation of transverse waves.

7.1 Describing progressive waves (Syllabus 7.1.1, 7.1.2, 7.1.4–7.1.7, 7.2.2)

7.2 Looking at longitudinal waves (Syllabus 7.1.1, 7.1.3, 7.1.5, 7.2.1, 7.2.2)

7.3 The Doppler effect for sound waves (Syllabus 7.3.1, 7.3.2)

7.4 The electromagnetic spectrum (Syllabus 7.1.5, 7.1.6, 7.4.1–7.4.3)

7.5 Polarisation (Syllabus 7.5.1, 7.5.2)

7.1 Describing progressive waves

WHAT IS A PROGRESSIVE WAVE?

A **progressive wave** is a periodic disturbance that transfers energy without transferring matter. This periodic disturbance is usually the regular vibration, or **oscillation**, of particles of the medium (material) through which the wave travels.

You can make a progressive wave by using a rope. Place the rope straight on the ground. Hold one end of the rope and move it rapidly from side to side. Figure 7.1 shows what happens.

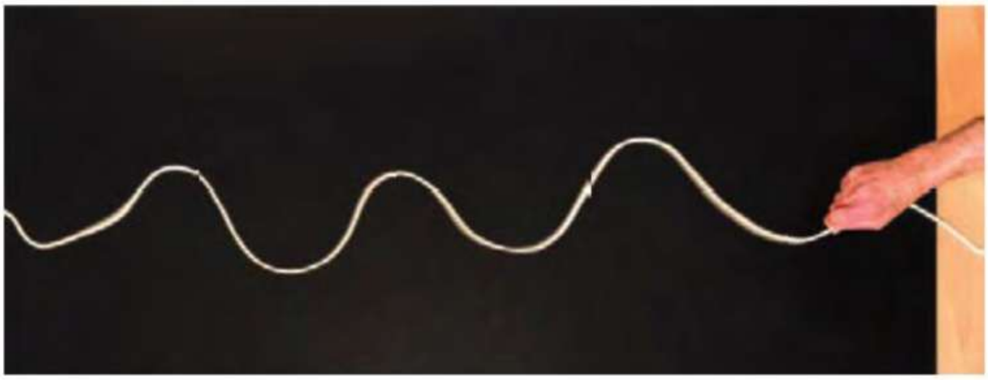

Figure 7.1 A progressive wave on a rope.

The wave in Figure 7.1 moves from your hand towards the other end of the rope, but the rope itself does not move this way. If you attached some small object to the other end of the rope, the wave would make it move. You are using the wave to transfer energy, but not matter, from your hand to the object.

You could put a marker, such as a piece of coloured thread, partway along the rope and observe how this marker moves. You would see it moving from side to side in the same manner that your hand does. This type of progressive wave is called a transverse wave.

TRANSVERSE WAVES

In a **transverse wave**, the oscillations are perpendicular to the direction of energy transfer.

Figure 7.2 shows particles in the rope at one instant in time and a short time later. When no wave is being produced the rope lies in a straight line. The particles are then in their equilibrium position. When the wave is produced by moving the hand from side to side, the particles (numbered 1 to 13) oscillate from side to side, passing though the equilibrium position. From the first instant of time to the second, particle 11 for example has moved towards the equilibrium position; point 8 has moved away from the equilibrium position. The particles do not move along the rope, only from side to side.

The distance of a particle from the equilibrium position at any point in time is called its **displacement**.

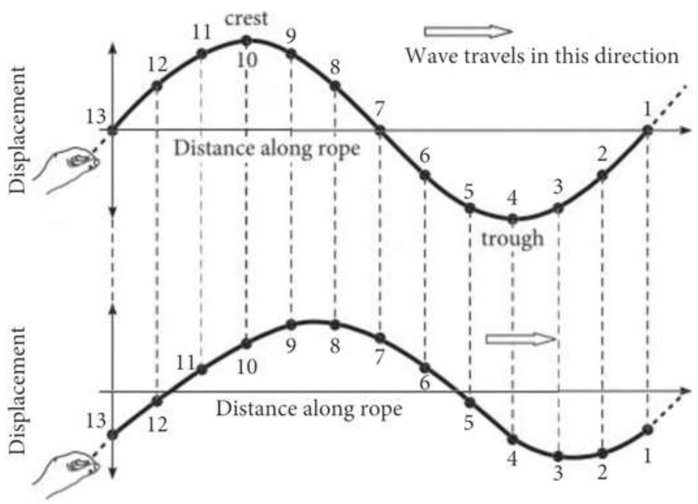

Figure 7.2 Particle displacement in a transverse progressive wave in a rope (top). The same wave is shown a short time later (bottom) as a result of the wave motion.

Waves in the vibrating strings of musical instruments, surface water waves, and also electromagnetic waves such as light and radio waves, are other examples of transverse waves.

WAVELENGTH AND AMPLITUDE OF WAVES

We can describe the form of all waves using the terms **wavelength** and **amplitude.** The wavelength and amplitude of a transverse wave are shown in Figure 7.3.

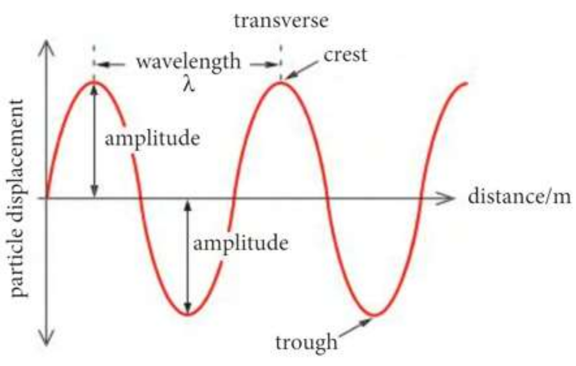

Figure 7.3 The wavelength and amplitude of a transverse wave.

> **Tip**
> On a diagram the amplitude of a wave is measured from the mid-point line of the wave to the top of a peak or the bottom of a trough. It is not the peak-to-trough distance.

The wavelength is given the symbol λ (the Greek letter lambda), and is the distance between any two adjacent equivalent points on the wave. Usually, you will measure this from one wave peak to the next, but it could be measured from any point on the wave to the next equivalent point.

The amplitude is the maximum displacement of the wave from the equilibrium position. The equilibrium position is the location of the particles when there is no disturbance due to the wave. In the rope in Figure 7.1, the equilibrium position of the rope is the straight line where it lay before being moved.

Figure 7.4 A disturbance in water causes waves.

The equilibrium position of the water waves in Figure 7.4 is the flat surface of the water before the disturbance that caused the waves.

1. Make a simple sketch of a wave. On your sketch
   (a) mark the direction of travel of the wave
   (b) label the wavelength, using the symbol λ
   (c) label the amplitude
2. Estimate the wavelength of the wave on the rope in Figure 7.1.
3. Calculate the number of wavelengths in 1.00 m for each of these waves.
   (a) a wave on a string of wavelength 50.0 cm
   (b) a water wave of wavelength 28.0 mm
   (c) light of wavelength 660 nm
   (d) microwaves of wavelength 870 μm

Link

You may want to look back to Chapter 1 to remind yourself of the unit prefixes micro (μ), nano (n) and pico (p) and their values in powers of 10. These prefixes are often used to quantify wavelengths of electromagnetic waves.

SPEED, FREQUENCY AND PERIOD OF WAVES

The speed of a wave is defined in the same way as the speed of any other object.

$$\text{speed (m s}^{-1}) = \frac{\text{distance convered (m)}}{\text{time (s)}}$$

$$v = \frac{s}{t}$$

The speeds of different types of waves vary considerably and also depend on the medium they are passing through (see Table 7.1). The speed of electromagnetic waves in a vacuum is one of the fundamental constants in the Universe. The speed of electromagnetic waves in a vacuum is a special value of speed, often denoted by c rather than v. This is the maximum possible speed in the Universe.

Link

You will learn more about sound waves in Topic 7.2.

wave and medium	speed / m s^{-1}
water waves caused by wind	2.5–15
sound waves in air at sea level at 20 °C	343
sound waves in sea water at 4 m depth at 20 °C	1500
electromagnetic waves in optical (fibre optic) cable	2.00×10^8
electromagnetic waves in air or vacuum*	3.00×10^8

Table 7.1 The speeds of some different waves

*The speed of electromagnetic waves in air and in a vacuum is actually slightly different, but the difference is so small that it does not affect the value at 3 significant figures.

Link

You may wish to look back at Chapter 1 because the prefixes kilo (k), mega (M), giga (G) and tera (T) are often used with frequency, for example MHz. It would be helpful to recall the values of these prefixes in powers of 10.

The speed of a wave is affected by the medium in which it travels (see Table 7.1). Electromagnetic waves travel slower when they pass through a material such as water or glass compared with their speed in a vacuum. Sound waves travel at a different speed in different materials.

The speed of a wave and its frequency, f, are related.

The **frequency** of a wave is defined as the number of complete oscillations per unit time. The unit of frequency is s^{-1} or hertz, Hz.

The frequency of a wave is proportional to the speed of the wave when all other factors are constant. As the wave travels faster, more complete waves pass a fixed point per second. So we can write

$$f \propto v \qquad (1)$$

For a wave moving at a particular speed, the longer its wavelength, the fewer complete waves will pass a point each second. The wavelength will be inversely proportional to the frequency. We can write

$$f \propto \frac{1}{\lambda} \qquad (2)$$

As no other factor influences frequency, the constant of proportionality in equation 1 is $\frac{1}{\lambda}$ and the constant of proportionality in equation 2 is v. Therefore

speed of a wave = frequency × wavelength

$$v = f\lambda$$

This equation is referred to as the **wave equation**.

The speed of electromagnetic waves in air or vacuum (see Table 7.1) is the same for all frequencies. The speed of sound in air, at a particular temperature and air density, is the same for all frequencies. Both electromagnetic waves and sound change speed in different media, but their frequency is unaltered (since this depends on the wave source), so their wavelengths will change according to the medium in which they travel.

For example, a sound of frequency 500 Hz in air travels at about 340 m s^{-1}. This wave has a wavelength of 340 m s^{-1} ÷ 500 s^{-1} = 0.68 m. The speed of sound increases approximately four times in water. As the frequency remains constant we can see from the wave equation that the wavelength must increase by the same factor.

The **period** of a wave, T, is the time taken for one complete oscillation to pass a fixed point. This is inversely related to frequency.

Consider a wave whose frequency is 10 Hz. This means 10 complete oscillations pass a fixed point per second. This also means that one oscillation will take $\frac{1}{10}$ s to pass a fixed point.

Therefore

$$f = \frac{1}{T}$$

Link

You may want to refer to the discussion of inverse proportionality in Topic 3.2.

Link

In fact the frequency of a source of waves can appear to change, if there is relative motion of the source and the observer. You will find out about this in Topic 7.3.

Tip

As $f = \frac{1}{T}$ then $T = \frac{1}{f}$

Worked example

The frequency of yellow light is 530 THz. Calculate the wavelength of this light in air. Give your answer in nm.

The speed of light in air = 3.00×10^{8} ms^{-1}

Answer

530 THz = 530×10^{12} Hz

1 Hz = 1 s^{-1}

$v = f\lambda$

$\lambda = \frac{v}{f}$

$\lambda = 3.00 \times 10^{8}\ \mathrm{ms^{-1}} \div 530 \times 10^{12}\ \mathrm{s^{-1}}$

$\lambda = 5.7 \times 10^{-7}$ m

1 nm $= 10^{-9}$ m

5.7×10^{-7} m $\div 10^{-9}$

$= 570$ nm

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4. Calculate the frequency of microwaves with a wavelength of 2.6 cm.
5. Some of the largest water waves to be caused by wind were observed by the US Navy in 1933 in the North Pacific Ocean. They had a peak-to-trough distance of 34 m, a wavelength of 342 m and a period of 14.8 s.
   For these waves, calculate
   (a) the amplitude
   (b) the frequency
   (c) the speed.

THE RIPPLE TANK

The ripple tank is a transparent glass or plastic tray that can be filled with water and used to show the wave behaviour of surface water ripples. Some have inbuilt projection screens (Figure 7.5) so that images of the water waves can easily be viewed. Alternatively a ripple tank can be placed on an overhead projector and the image of the water waves is focused on a large vertical screen.

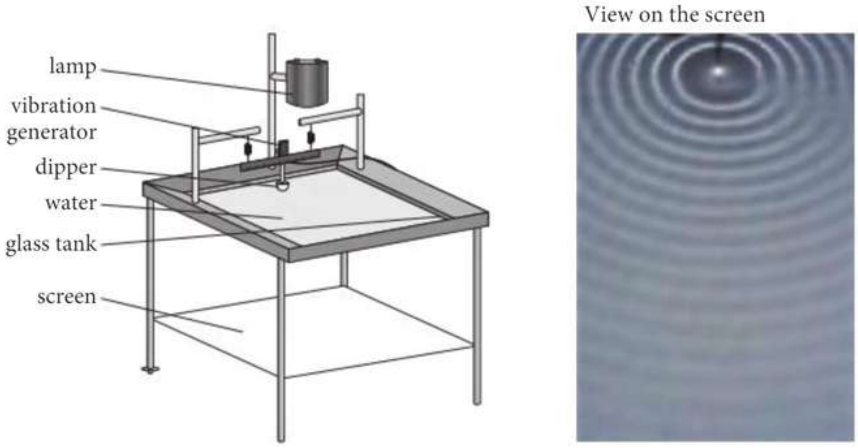


Figure 7.5 A ripple tank and the projected image of circular waves.

The ripple tank in Figure 7.5 uses a round 'dipper' that dips into the water and vibrates up and down to produce circular water waves. The dipper can be replaced with a rectangular 'paddle' to generate plane (straight) waves. The vibrations are generated using an electric motor.

Link

You will learn more about the uses of ripple tanks and the other wave patterns that they can produce in Chapter 8. You will also see how they can be used to investigate wave properties such as diffraction and interference.

Experimental skills 7.1: Variation in the speed of water waves with depth

If you watched water waves arriving on a gently sloping beach, you might notice that their speed changes. This is caused by the change in depth of the water. We can investigate the relationship between the speed of water waves and depth in the laboratory.

Apparatus

You will need:

- a plastic or metal tray minimum 30 cm long and with sides minimum 5 cm high
- a metre rule
- a ruler suitable for measuring the depth of water
- a source of water and container to pour water
- a stopwatch

Techniques

Measure the length of the tray. Put water into the tray so it has a depth of 10 mm.

Lift one end of the tray to a height of 1–2 cm above the desk or bench where you are working. Drop this end of the tray and watch the wave pass across the water. Start the stopwatch when the wave reflects off the opposite end of the tray.

Observe the resulting wave pass across the water. It should be reflected from each end of the tray to cross the length of the tray 5–7 times. Stop the stopwatch when the wave has made the largest number of observable passes possible. The speed of the wave does not depend on the height from which the tray is dropped.

Repeat this with water depths increasing by 5 mm each time up to a maximum of 35 mm.

Process the data to obtain a water speed for each depth.

QUESTIONS

P1. Explain what is meant by 'a ruler suitable for measuring the depth of water'.

P2. State two control variables for this investigation.

P3. **(a)** Explain why the water depth should be measured in different positions and a mean calculated.

(b) Describe how the mean water depth value could be made more accurate.

P4. Draw a suitable table for recording the results.

P5. A student recorded the following repeat results for a water depth of 10 mm. The times are in seconds for 6 passes forward and back across a tray 40 cm in length.

7.48 7.51 7.53 7.60 7.43

(a) Find the uncertainty in these time measurements and calculate the percentage uncertainty in the smallest value.

(b) Calculate the wave speed in m s^{-1} from these results.

P6. Theory suggests that the relationship between wave speed, v, and water depth, h, is $v = \sqrt{gh}$ where g = the acceleration due to gravity, 9.81 m s^{-2}.

(a) Explain why plotting a graph of v against h would not produce a straight line.

(b) The student carries out the investigation again and records their calculated values for mean wave speed in Table 7.2.

mean water depth, h / mm	wave speed, v / m s^{-1}				mean wave speed, v / m s^{-1}	v^2 / m^2 s^{-2}
10	0.30	0.32	0.31		0.31	0.096
15	0.38	0.33	0.37	0.39	0.38	0.14
20	0.43	0.42	0.44		0.43	0.18
25	0.46	0.49	0.49		0.48	
30	0.52	0.55	0.59	0.55		
35	0.58	0.56	0.57		0.57	

Table 7.2

(i) State the units of v^2 that are missing from Table 7.2.

(ii) State why the 0.33 m s^{-1} value for water depth 15 mm is circled.

(iii) Identify any anomalous result for water depth 30 mm and calculate the missing value of mean wave speed v.

(iv) Calculate the missing values of v^2.

(v) Use these results to plot a graph of v^2 against h and complete the graph with a straight line of best fit.

P7. (a) Calculate the gradient of the graph. What does the gradient represent?

(b) (i) Calculate the percentage uncertainty in the lowest wave speed.

(ii) Estimate the percentage uncertainty in the smallest value of h.

(iii) Hence suggest which quantity contributes the greatest uncertainty to the calculated value of g from this investigation.

A Level analysis

P8. (a) Plot a graph of v^2 against h and include error bars for h.

(b) Draw a straight line of best fit and the worst acceptable line. Label both lines clearly.

(c) From your graph determine a value for g and the percentage uncertainty in this value.

PHASE

Figure 7.6 represents two wavelengths of a transverse wave. Each of the numbered points on the wave is at a different position on the wave, with a different displacement. Points with the same number are equivalent points on the wave. For example, points labelled 8 are one wavelength apart; they have not only the same displacement but they are moving in the same direction. We say they are oscillating **in phase**.

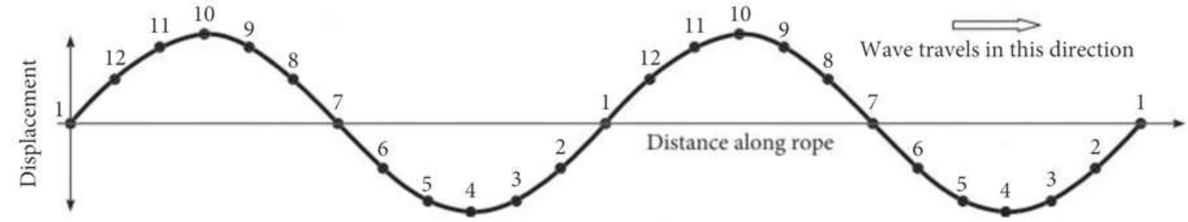


Figure 7.6 Displacement–distance for two wavelengths of a transverse progressive wave. Here the numbers indicate positions of equivalent phase.

The graph of displacement of particles on a transverse wave against distance travelled by the wave, for displacement zero at distance zero, has the form of the graph of sin θ against θ.

Similarly, if we plot a graph of the displacement of one particle against time, for displacement zero at time zero (Figure 7.7), this also has the form of the graph of sin θ against θ.

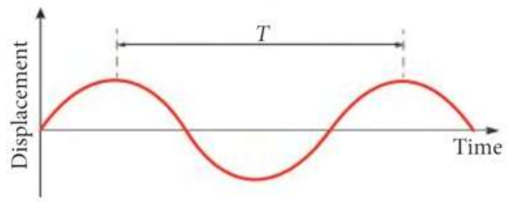


Figure 7.7 Displacement–time for one point on a progressive wave. The period T is the time interval between two successive equivalent times in the oscillation

The graph of sin θ against θ is shown in Figure 7.8.

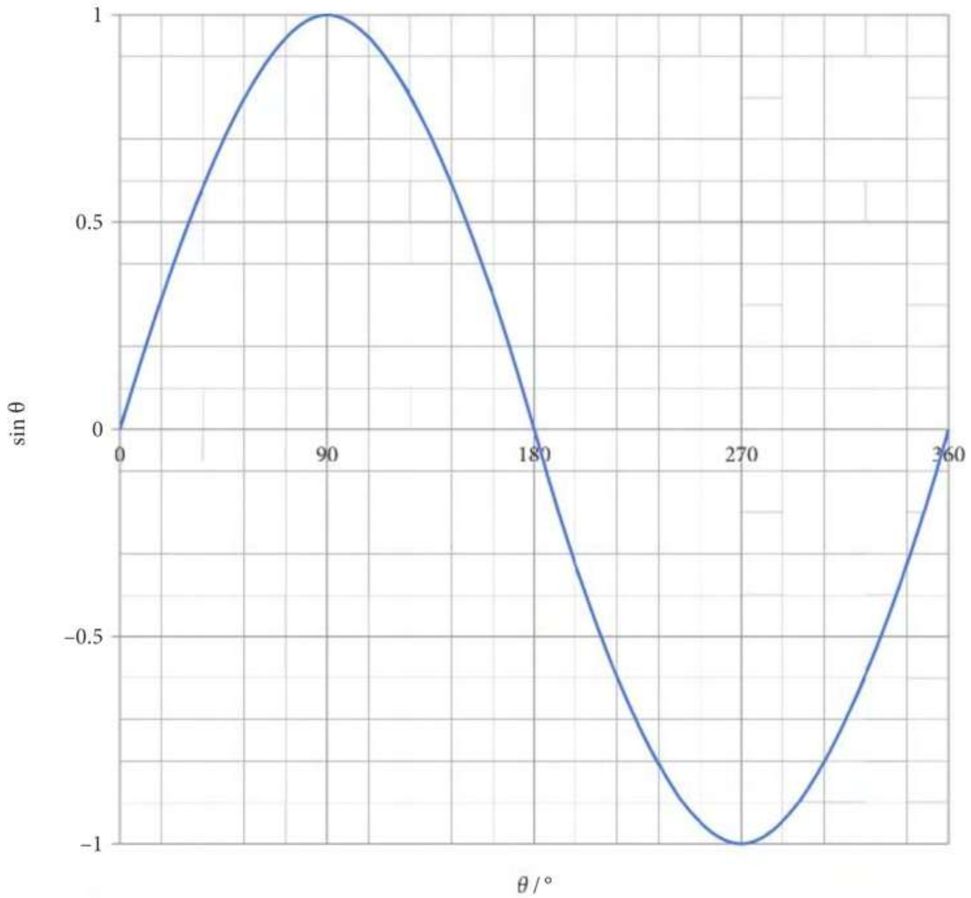


Figure 7.8 A sine curve for angles in degrees.

This means that, instead of using arbitrary position numbers 1 to 12 as in Figure 7.6, we can use an angle from 0 to 360° to refer to a point on a wave (a point in space or a point in time). This is called the **phase angle**.

A phase angle of 360° is equivalent to a distance of one wavelength or a time interval of one period.

Phase differences in degrees (and at A Level, radians)

We can use phase angles to compare the progression of two waves. The two waves in Figure 7.9, for example, have a **phase difference** of 90°. There is one-quarter of a period time lag between them; one-quarter of 360° is 90°. Phase difference is the difference in phase angle between two waves at the same time.

The red wave is leading the green wave. We can see this by comparing the time, t, values for equivalent points. We can tell the red wave is one-quarter of a period ahead of the green one because the green wave matches the red one 2 seconds after it.

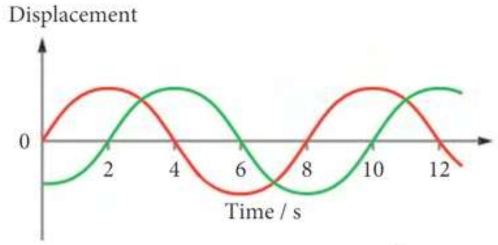


Figure 7.9 Displacement–time for two waves with a phase difference of $\frac{1}{4}$ T.

At A Level, we often use degrees as the unit for measuring angles, but we can also use **radians** as the unit for angles. Phase angles and phase differences are commonly expressed in radians.

Tip

Only change your calculator setting from degrees to radians when you are using the trigonometry functions, sin, cos, tan, $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$. Set it to degrees to evaluate, for example, sin 40° and to radians to evaluate, for example sin 0.10 rad.

Using the unit radians, angles can be expressed in terms of the constant π (pi).

$$360° = 2\pi \text{ radians}$$

- If two waves have a phase difference of 0 or 2π (or 4π, …) they are in phase.
- If two waves have a phase difference of π (or 3π, …) they are **in antiphase**.

Some significant phase differences between waves and their descriptions are given in Table 7.3.

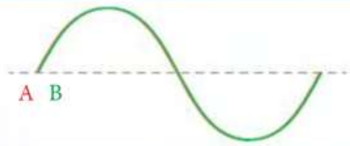
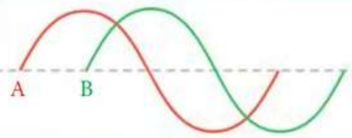
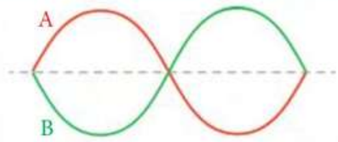
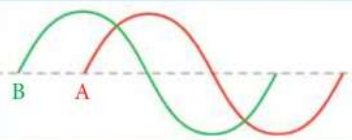
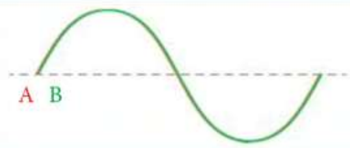
phase difference between A and B / degrees	/ radians	description	appearance of A and B
0	0	in phase	
90	$\frac{\pi}{2}$	-	
180	π	in antiphase	
270	$\frac{3\pi}{2}$	-	
360	2π	in phase	

Table 7.3 Phase differences between two waves A and B.

A Level only

6. Sketch a graph of displacement again distance for a transverse wave to show at least two wavelengths.
   Mark on your sketch the phase angles
   (a) 0
   (b) $\frac{\pi}{2}$ radians
   (c) π radians
   (d) $\frac{3\pi}{2}$ radians
   (e) 2π radians
7. Which of these pairs of phase differences are the same?
   A 0 and $\frac{\pi}{2}$ radians
   B 0 and π radians
   C 0 and $\frac{3\pi}{2}$ radians
   D 0 and 2π radians

8. Consider the dashed lines in the wave diagrams in Table 7.3 representing the time axis of a displacement-time graph.
   State which wave is leading when the phase difference is 270°.
9. Sketch two waves, A and B, that have equal amplitude and equal wavelength, but are π radians out of phase.

Link

It may be helpful to look back at Chapter 5. In particular, the relationship between energy and power and also how kinetic energy is related to speed.

WAVE INTENSITY

Intensity is a measure of the rate of transfer of energy by a wave to unit area that is perpendicular to the direction of travel of the wave.

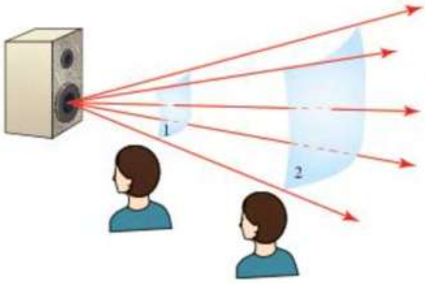


Figure 7.10 The intensity of the sound waves is greater at position 1 as there is more energy transferred in unit time to unit area.

In Figure 7.10, the person will hear a louder sound at position 1 than at position 2, because the sound intensity is greater at position 1.
The unit of intensity is W m^{-2}.

$$\text{intensity (W m}^{-2}) = \text{power (W)} \div \text{area (m}^2)$$

When a wave passes through a medium, it is the kinetic energy of the oscillating particles that is transferred. As kinetic energy is given by $\frac{1}{2}mv^2$, the kinetic energy of the particles is proportional to the square of their speed.

The maximum speed of the oscillating particles is proportional to their amplitude of oscillation, as long as the frequency does not change. If the amplitude doubles, then the particles have to travel twice as far in the same time, so their speed must double.

This gives us the relation

$$\text{intensity} \propto (\text{amplitude})^2$$
$$I \propto A^2$$

10. A wave initially has amplitude H and intensity Y.
    What will be the intensity when the amplitude is increased to $3H$?
    A $\frac{1}{3}Y$
    B $3Y$
    C $9Y$
    D $27Y$
11. The intensity of a wave decreases by a factor of 0.5. State the factor by which the amplitude decreases.

Key ideas

- A progressive wave transfers energy without the transfer of matter.
- In a transverse wave, the vibrations are perpendicular to the direction of wave travel. This is observed as peaks and troughs.
- Examples of transverse waves are those in the electromagnetic spectrum, those in stretched strings, and water waves.
- All progressive waves can be described by their speed, amplitude, wavelength, and frequency.
- wave speed = frequency × wavelength
- Positions on a wave can be referred to as phase angles, using degrees or radians.
- The difference in progression of two waves of equal wavelength can be stated as phase difference using degrees or radians.
- Intensity is the rate of energy transfer per unit area perpendicular to the wave travel.
- intensity $\propto$ (amplitude)^2

7.2 Looking at longitudinal waves

In Topic 7.1 we described the oscillations in transverse waves and saw how to make measurements from their graphical representation. In this topic we will describe the particle motion in sound waves and look at how to determine the frequency, wavelength and amplitude of sound waves.

LONGITUDINAL WAVES

Sound waves travel differently from transverse waves. In a sound wave, the particles of the medium oscillate in a direction parallel to the direction of energy transfer. This type of progressive wave is called a **longitudinal wave**.

The terms amplitude, speed, wavelength, frequency, phase and intensity that you met in Topic 7.1 can all be used to describe longitudinal waves, as well as transverse waves. The wave equation $v = f\lambda$ also applies.

You can make a longitudinal wave using a "slinky" spring. Place the spring straight on the ground. Fix, or ask someone to hold, one end of the spring. Then push and pull the other end rapidly. You will see waves of compression (where the turns of the coil are pushed close together) moving along the spring, but there is no motion from side to side. This is shown in Figure 7.11.

The regions of compression in the spring are a model of regions of compression in a sound wave in air. These are the regions where the molecules in the air are closer together, so the air pressure is higher.

Between the regions of compression are regions of rarefaction. The turns of the spring are further apart, modelling regions of low air pressure in a sound wave in air.

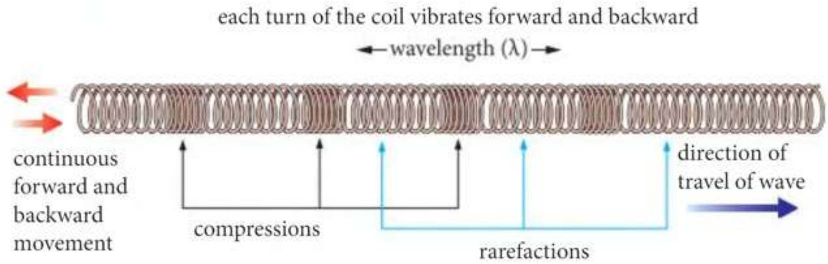


Figure 7.11 A longitudinal wave in a spring.

The wavelength of a longitudinal wave, as shown in Figure 7.11, is the distance between adjacent equivalent points on the wave. This is most easily measured between the centre of one compression and the centre of the next compression (or between two successive rarefactions, although the centres of these may be more difficult to determine).

Sound waves travel as longitudinal pressure waves through gases, liquids and solids. The denser the medium, the faster the sound wave travels. If we consider a sound wave moving through air, we can show how the atmospheric pressure varies along the wave (Figure 7.12).

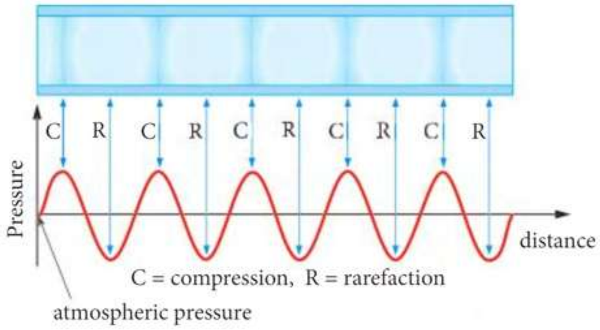


Figure 7.12 The top part represents the regions of compression of a longitudinal sound wave in air. The lower part shows how the atmospheric pressure varies with distance. The variation in atmospheric pressure of many sound waves is only about 0.03%.

A displacement–time graph for an oscillating particle in a longitudinal wave will be a sine curve, similar to the displacement-time graph of a transverse wave (Figure 7.7). The particles oscillate about their equilibrium (undisturbed) positions. But in a longitudinal wave the displacements are parallel to the direction of wave motion. The forward-and-back motion is represented by up and down on the graph. Maximum displacement corresponds to a compression, minimum displacement to a rarefaction.

The amplitude of a longitudinal wave is not easy to envisage. As for a transverse wave, the amplitude is the maximum displacement of the particles.

A longitudinal wave with greater amplitude will have higher magnitude compressions. The coils of the spring will be pushed closer together. The molecules in air will be pushed closer together so the air pressure will be greater. The magnitude of the rarefactions also increases with amplitude. The coils of the spring will be pulled further apart. Molecules in air will be further apart causing lower air pressure.

As we saw in Topic 7.1, the intensity of a wave depends on the square of its amplitude. A sound wave of higher amplitude, and hence higher intensity, produces a louder sound.

12. A person playing a guitar is performing for a television broadcast.
Classify each of these waves in parts (a) to (e) as transverse or longitudinal.
- **(a)** Vibration on the guitar string.
- **(b)** Sound from the guitar travelling through air to the microphone.
- **(c)** Radio waves from the transmitter to a satellite.
- **(d)** Sound from a loudspeaker in a person's home to their ear through air.
- **(e)** Sound from the loudspeaker through the wall into the next room.

13. Write the waves in question 12 (b) (c) and (e) in order of speed from slowest to fastest.

Link

You may want to look back to Table 7.1 in Topic 7.1.

14. Figure 7.13 shows the variation in displacement of a molecule in air with time during the passing of a sound wave.

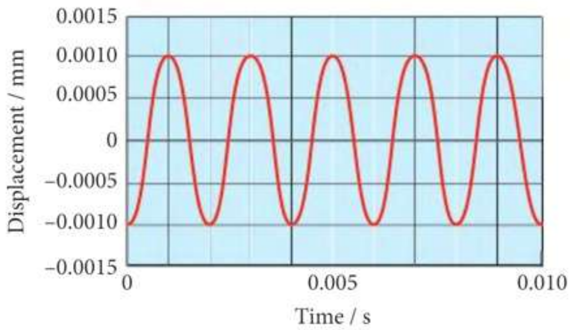


Figure 7.13

(a) State whether the wave represented in Figure 7.13 is transverse or longitudinal.
(b) Use Figure 7.13 to determine the frequency of the wave.
(c) The speed of the wave in Figure 7.13 is 340 m s^{-1}. Calculate its wavelength.
(d) Use Figure 7.13 to estimate the maximum speed of the molecule due to the wave.

15. (a) State two similarities between transverse and longitudinal progressive waves.
(b) State two differences between transverse and longitudinal progressive waves.

OBTAINING A GRAPHICAL REPRESENTATION OF SOUND WAVES

The longitudinal nature of sound waves makes this type of wave difficult to draw or to take measurements from directly. However, we can use a microphone or a signal generator, together with a cathode-ray oscilloscope (CRO), to display sound waves in a transverse form and then take measurements from the display.

Figure 7.14 A microphone converts sound waves to electrical signals.

The microphone

A microphone is a device which converts sound waves into varying electrical signals. The pressure wave is converted into a varying voltage. Stage performers, public speakers or musicians often use microphones. In this case, the microphones are connected to amplifiers, which make the sound louder.

Microphones are manufactured in a range of different shapes and sizes. A typical microphone is shown is Figure 7.14.

Figure 7.15 A signal generator.

The signal generator

Sounds produced by musical instruments contain more than one frequency, even when just one single note is played. This is how you can tell the difference between the same note played on different types of musical instruments. To produce sound waves that contain only one frequency, we use a **signal generator** and a loudspeaker.

There are different types of signal generator. All of them have an adjustable frequency output and some way to control the amplitude of the output wave. If a signal generator can produce different wave forms, such as square, triangular or sine, then the sine wave must be used to produce sounds.

The cathode-ray oscilloscope

A cathode-ray oscilloscope, or simply **oscilloscope** or **CRO**, is a device for displaying electrical signals on a screen. There are many different types of oscilloscope. One type is shown in Figure 7.16.

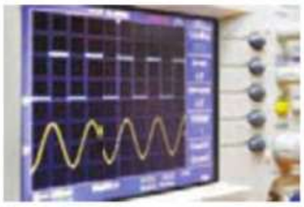
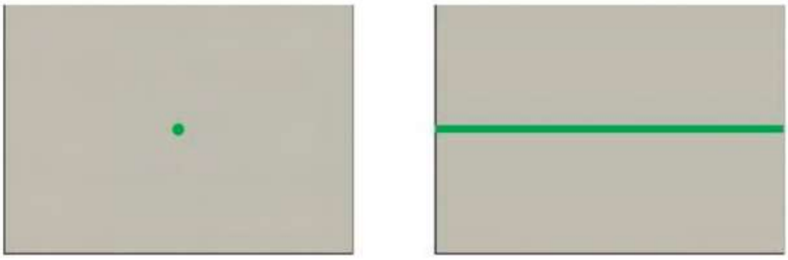


Figure 7.16 An oscilloscope, or CRO, displaying electrical signals as square waves and sine waves.

There are many controls on an oscilloscope, but the most important are the time-base and the *y*-gain.

- The time-base controls how quickly the trace moves across the screen and needs to be adjusted so that your wave is displayed in a manner that you can see clearly.
- The *y*-gain, or simply gain, controls the height of the displayed wave.

When an oscilloscope has no input and the time-base is off, a single spot is seen on the screen. The time-base is an applied varying voltage that makes the spot move across the screen from left to right. A time-base voltage frequency of 20 Hz means the spot moves across the screen 20 times per second. When the spot moves quickly it is seen as a straight line, or 'trace', as shown in Figure 7.17.

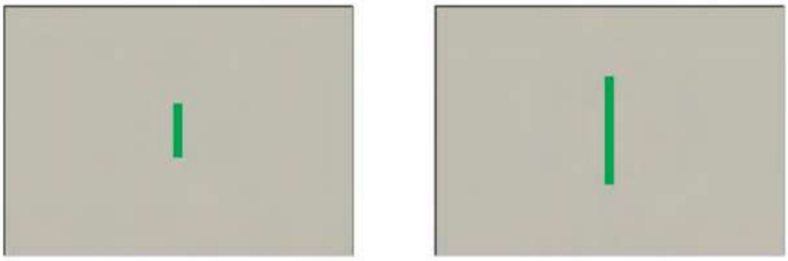


time-base off　　time-base on

Figure 7.17 The effect of the oscilloscope's time-base.

When an electrical input is connected to the oscilloscope, the input voltage causes the spot to be displaced in the vertical, or *y* direction. The electrical input might be from a microphone or a signal generator.

The *y*-gain control provides an adjustment so that the vertical displacement of the spot can be seen clearly, yet remain within the visible part of the screen. Figure 7.18 shows the appearance of a sine wave input with the time-base off and two different *y*-gain settings.



y-gain low　　*y*-gain increased

Figure 7.18 The effect of the oscilloscope's y gain on the same input with the time-base off.

We can make comparative measurements of inputs with an oscilloscope by using the grid on the oscilloscope screen. The time-base setting (Figure 7.19a) is stated in milliseconds per division or microseconds per division (ms div^{-1} or μs div^{-1}). The *y*-gain setting (Figure 7.19b) is stated in volts per division or millivolts per division (V div^{-1} or mV div^{-1}).

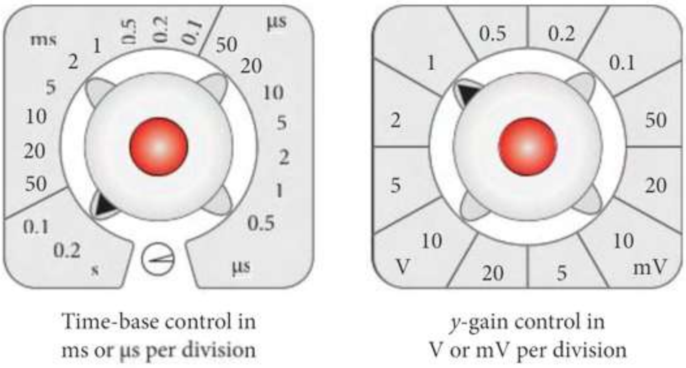


Figure 7.19 Time-base and y-gain controls on an oscilloscope

Figure 7.20 shows the effects of the time-base and the y-gain together on the same sine wave input voltage. The oscilloscope display now resembles a graph of voltage against time. Changing the time-base and y-gain settings with the same input has the same effect as changing the scale on a graph.

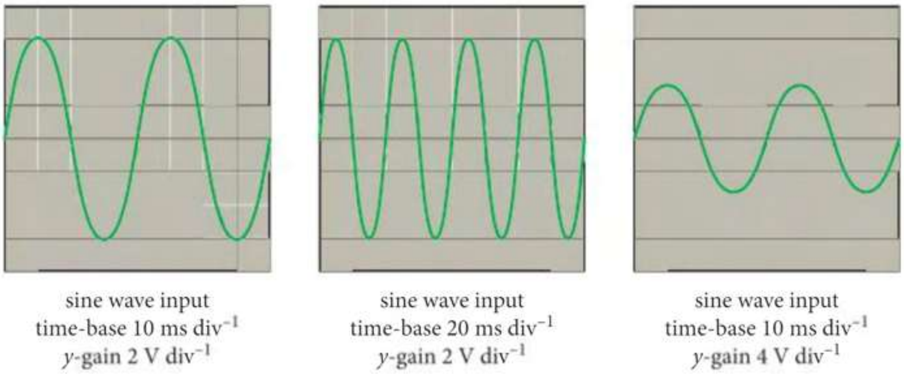


Figure 7.20 The effects of time-base and y-gain settings on the same sine wave input

If the time-base and y-gain settings on an oscilloscope are kept the same, comparisons can be made between the frequencies and amplitudes of different waves.

16. Figure 7.21 shows oscilloscope screen displays for various outputs of a signal generator.

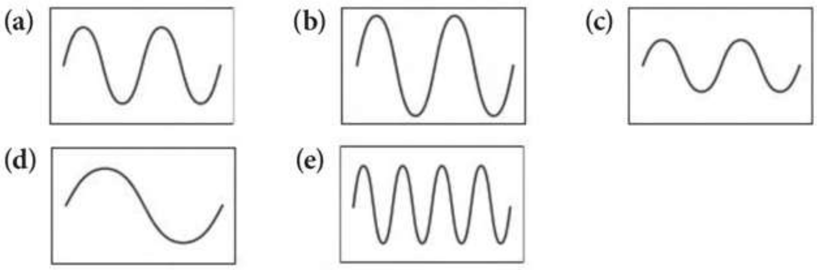


Figure 7.21

The signal generator is first set up and connected to the oscilloscope as shown in Figure 7.21 (a).

Four separate adjustments are made to the signal generator as shown in Figure 7.21 (b) (c) (d) and (e).

The time-base and y-gain settings on the oscilloscope are not changed.

Describe what has been changed on the signal generator to produce each of the outputs shown in Figure 7.21 (b) to (e). For each one, compare the frequency and amplitude with (a).

A loudspeaker

A loudspeaker, or just speaker, does the opposite of a microphone. It converts electrical signals into sound waves. A cone made from paper or plastic vibrates with the same frequency as the input electrical wave. The vibration of the cone produces sound waves in air.

When using a loudspeaker, the output level from the signal generator must be adjusted so that the sound from the loudspeaker is not distorted. (When the volume is too high the loud speaker produces a distorted sound wave.) If the sound is distorted, it will be more difficult to take measurements from the displayed wave.

USING AN OSCILLOSCOPE TO FIND THE FREQUENCY OF SOUND

We can connect a microphone to an oscilloscope and produce a graphical display that shows the characteristics of a sound.

The microphone converts a sound wave into a varying voltage, which can then be displayed on the oscilloscope. The screen displays a voltage-time graph which is a representation of the sound wave. The frequency of the wave cannot be directly measured, but the period of the wave can be worked out from the oscilloscope's time-base setting (the time per division on the screen), and then the frequency calculated from the period.

The equipment is set up as shown in Figure 7.22.

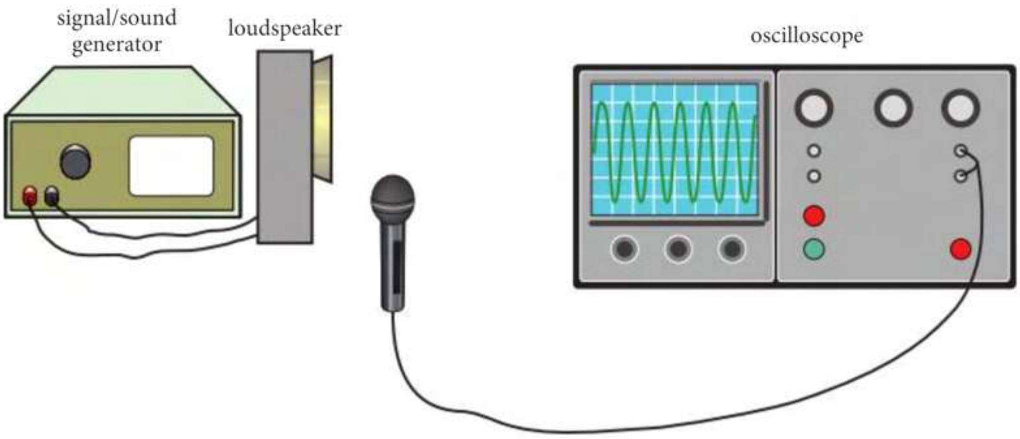


Figure 7.22 Using an oscilloscope to display characteristics of a sound wave.

The time-base setting may need to be adjusted so that only two or three peaks are displayed, as in Figure 7.23, to allow easy measurement of the period.

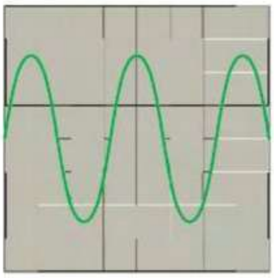


Figure 7.23 An oscilloscope display representing a sound wave. A pure note of a single frequency is a sine wave

> **Tip**
> The distance between two peaks on the oscilloscope display gives the period of the wave, not the wavelength of the wave.

> **Tip**
> The amplitude of the displayed sine wave can be determined from the y-gain setting. This is the amplitude of the input electrical signal in volts and not a direct measurement of the amplitude of the sound wave.

Worked example

The oscilloscope display in Figure 7.23 is produced when a microphone is connected to the oscilloscope. The microphone is close to a loudspeaker which is connected to a signal generator. The time-base setting is 0.2 ms div^{-1}.

(a) Determine the frequency of the sound.

(b) Taking the speed of sound in air to be 340 m s^{-1}, calculate the wavelength of the sound wave in the air between the loudspeaker and microphone.

Answer

(a) Two complete waves span exactly 10 horizontal scale divisions, so the period of the wave is $\frac{1}{2} \times 0.2\text{ ms} \times 10 = 1.0\text{ ms}$ or 1.0×10^{-3} s.

$f = 1 \div T \Rightarrow f = 1 \div 1.0 \times 10^{-3}\text{ s} = 1000\text{ Hz}$

(b) $v = f\lambda$

$\lambda = v \div f = 340\text{ m s}^{-1} \div 1000\text{ s}^{-1} = 0.34\text{ m}$

17. Figure 7.24 shows an oscilloscope screen. The input is direct from a signal generator. The time-base is set to 2 ms div^{-1} and the y-gain is 0.5 V div^{-1}.

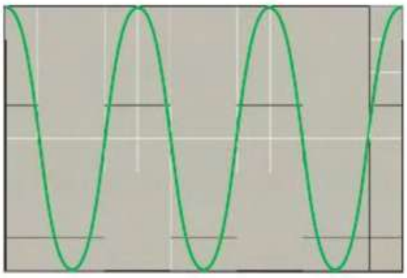


Figure 7.24

Calculate:

(a) the amplitude of the input signal in V

(b) the frequency of the wave in Hz.

18. An oscilloscope time-base is set to 10 ms cm^{-1}. Eight complete waves cover a horizontal distance of 5 cm on the screen. Calculate the frequency of the wave.

19. The output of a signal generator has a frequency of 500 Hz and an amplitude of 0.20 V. An oscilloscope is connected to the signal generator and the display is shown in Figure 7.25.

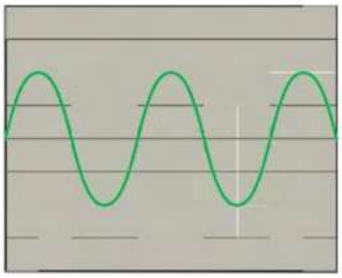


Figure 7.25

Calculate:

(a) the time-base setting on the oscilloscope in ms div^{-1}

(b) the y-gain setting on the oscilloscope in V div^{-1}

★**20.** A dual-trace oscilloscope is connected to two signal generators. Both signal generators are set to the same frequency. The oscilloscope time-base for each input is set to 7.5 μs div^{-1}. The oscilloscope display is shown in Figure 7.26.

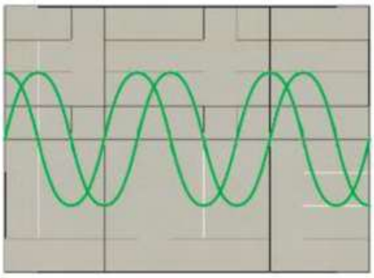


Figure 7.26

Calculate:

(a) the frequency of the inputs

(b) the phase difference between the inputs

(i) in degrees

(ii) in radians in terms of π.

Key ideas

- Sound waves are longitudinal pressure waves.
- In a longitudinal wave, the vibrations are parallel to the direction of wave travel. This is observed as regions of compression and rarefaction.
- A microphone or signal generator connected to an oscilloscope can be used to produce a graphical representation of a sound wave.
- The oscilloscope display shows a voltage–time graph from which the period and frequency of the original sound wave can be determined.

Experimental skills 7.2: Determination of the speed of sound in air.

When you hear a sound, you can usually tell whether it comes from the left or from the right. How does this work? As we have two ears, the sound will not arrive at both ears at exactly the same time unless the sound comes directly from the front or rear. A sound coming from the left, for example, will arrive at your left ear first. The human brain detects this difference and hence you know where the sound comes from. But what is this time interval? If we determine the speed of sound we can calculate the time interval.

Apparatus

You will need:

- a signal generator set to produce a single pulse of sound, or other means of producing a short sound
- a loudspeaker, with wires for connection, if using a signal generator
- two identical microphones (sensitive loudspeakers will function as microphones) with long wires for connections
- stands for the microphones
- an oscilloscope
- a tape measure or metre rule

Techniques

Set up the apparatus as shown in Figure 7.27.

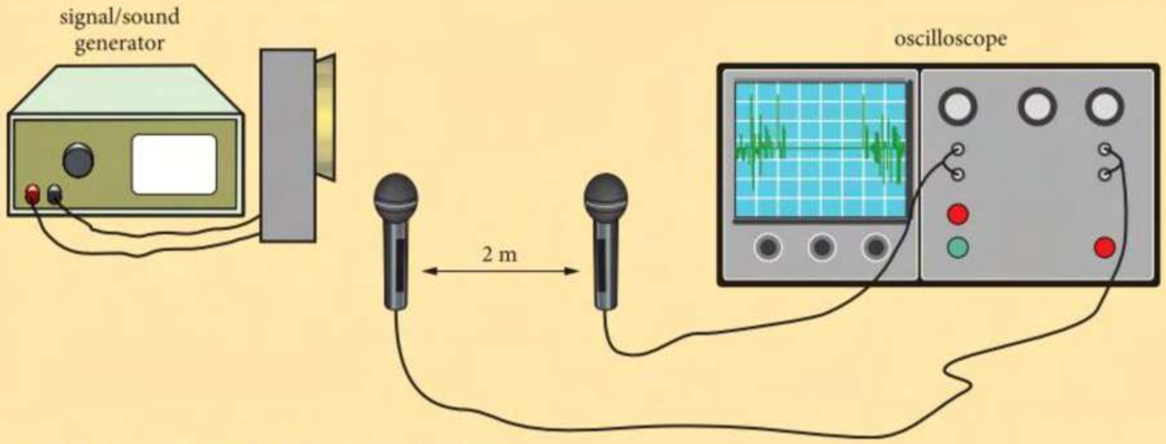


Figure 7.27 Determining the speed of sound in air.

Some oscilloscopes only have one input (or channel), so it is not possible to connect the microphones as shown in Figure 7.27. In this case, both microphones can be connected in parallel across the same input.

The sound source and the two microphones should be aligned so that they are along the same straight line.

Set up the oscilloscope so that the wave form from your sound source is showing clearly on the screen. Adjust the time-base so that the time interval between the sound arriving at each microphone is apparent and possible to measure. Figure 7.27 shows how the trace may appear. It may be possible to set the oscilloscope's trigger function so that the wave form remains stationary on the screen, to make the time measurement easier.

QUESTIONS

P1. Explain why the microphones should not be positioned close to a wall or glass window.

P2. The suggested distance between the microphones for this investigation is 2 m. Suggest one advantage and one disadvantage of using a larger distance between the microphones.

P3. **(a)** Estimate the expected time interval between the sound arriving at the first and second microphones when their separation is 2.0 m.

(b) Explain why the distance between the loudspeaker and the first microphone does not affect the separation between the sounds as seen on the oscilloscope screen.

(c) Suggest why the first microphone should not be positioned very close to the loudspeaker.

P4. The time intervals between the sounds are measured for a range of different distances between the two microphones.

(a) State the independent and dependent variables that should be measured in this investigation in order to determine the speed of sound from a graph.

(b) Explain how these variables should be plotted on a graph to determine the speed of sound.

(c) Explain why determining the speed of sound from a graph is preferable to taking measurements from one fixed setup and performing a single calculation.

7.3 The Doppler effect for sound waves

Think of a stationary car sounding its horn. When you stand beside it you hear the horn at a certain pitch. We detect the pitch of a sound by its frequency, which is the number of complete waves reaching your ear per second.

Figure 7.28 shows sound waves moving towards an observer from a stationary car. The curved lines represent the regions of compression in the air. They are curved because the waves are spreading out equally in all directions from the horn.

The distance between the compressions is the wavelength. If the wavelength does not change, the number of compressions that reach the observer per second is constant and is determined by the speed of the wave.

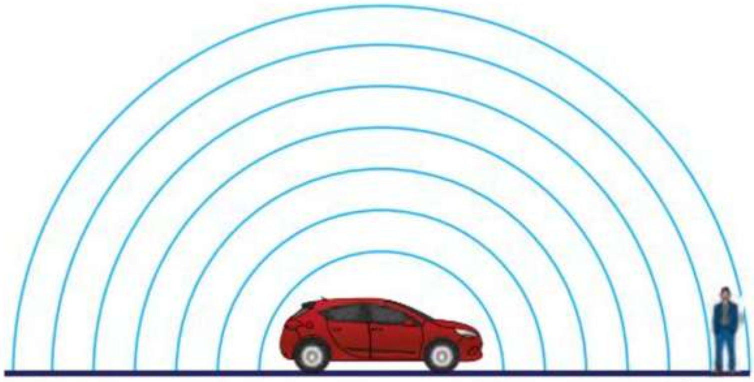


Figure 7.28 Sound waves from a stationary car's horn reaching an observer.

Now think what will happen if the car is moving quickly towards the observer, sounding the same horn (Figure 7.29). The frequency of the sound waves leaving the car will be the same as before, as the horn is still making the same sound. The speed of the sound waves in air does not change, because that is determined by properties of the air.

However, the wavelength of the sound waves between the car and the observer becomes shorter. More compressions will reach the observer per second. This is detected as an increase in pitch. The observed frequency is therefore greater than the source frequency.

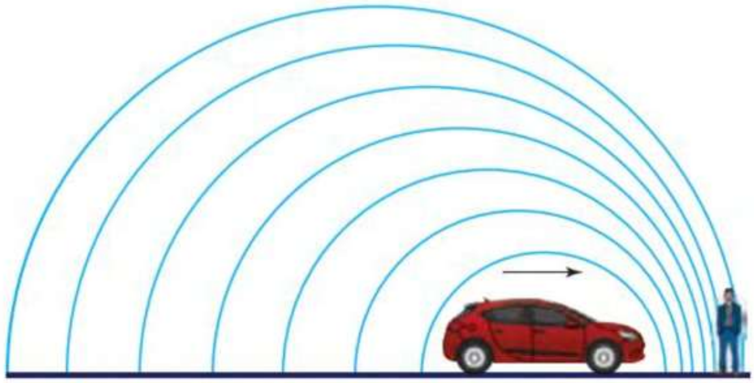


Figure 7.29 Sound waves from a car that is moving quickly towards an observer. There is an apparent increase in frequency, and hence pitch, of the sound to the observer.

The opposite is true when the car is moving quickly away from the observer. In this case, the compressions will arrive less frequently, so the pitch heard by the observer is lower (Figure 7.30).

This apparent change in wave frequency with relative motion of a source and an observer is called the **Doppler effect**.

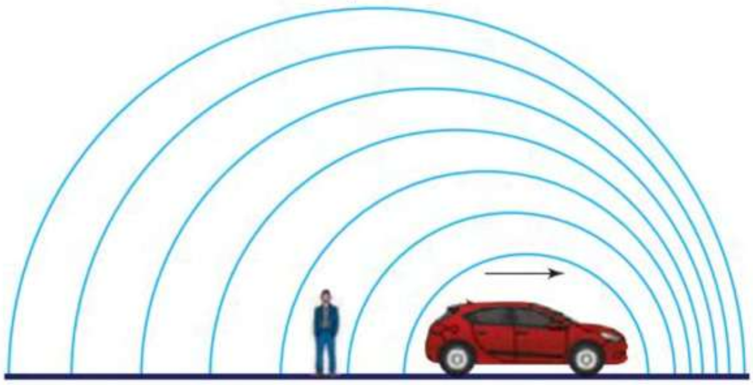


Figure 7.30 Sound waves from a car that is moving quickly away from an observer. There is an apparent decrease in frequency, and hence pitch, of the sound to the observer.

CALCULATING THE OBSERVED FREQUENCY

From the wave equation $v = f\lambda$, we know that the frequency of the waves is inversely proportional to the wavelength, for constant wave speed v.

We can find an expression for the observed wavelength, λ_o. Think of the case where the source is moving towards the observer.

A compression that leaves the source will travel a distance v m in one second.

The source travels towards the observer with speed v_S, so the source moves a distance v_S m in one second.

Another compression, which leaves the source one second after the first compression, will therefore be a distance $(v - v_S)$ m behind the first compression.

The number of compressions leaving the source in one second is the source frequency, f_S.

So the distance $(v - v_S)$ m contains f_S wavelengths. That is,

$v - v_S = f_S \times \lambda_o$

Therefore, the observed wavelength λ_o is given by

$$\lambda_o = \frac{(v - v_S)}{f_S}$$

The observed frequency f_o is then given by $f_o = v \div \lambda_o$:

$$f_o = \frac{f_S v}{(v - v_S)}$$

A similar exercise can be shown to give the observed frequency when the source is moving away from the observer:

$$f_o = \frac{f_S v}{v + v_S}$$

A general equation that gives the observed frequency when the source is moving either towards or away from the observer is

$$f_o = \frac{f_S v}{v \pm v_S}$$

Tip

When the distance between the source and the observer is increasing, the $\pm$ sign in the equation is positive. When the distance is decreasing, the sign is negative.

where the ± sign becomes − when the source is moving towards the observer and + when moving away.

The Doppler effect is only noticeable if the speed of the source relative to the observer is a significant proportion of the wave speed. The effect was first studied in sound waves, but it is also evident in the electromagnetic radiation from distant objects in space that are travelling away from us at extremely high speed.

Link

Chapter 25 will describe how Doppler shifts in the frequency of electromagnetic light from distant galaxies provide important information about the Universe and its evolution.

21. Calculate f_o as a percentage of f_S for:

(a) $v_S = 0.02v$ away from an observer

(b) $v_S = 0.2v$ away from an observer

Worked example

A police car is travelling on a straight road at a constant speed of 35 m s⁻¹. It has a siren that emits a sound of frequency 1200 Hz. The police car passes a person who is standing close to the road. The speed of sound in air is 340 m s⁻¹.

Calculate the frequency that the person observes when

(a) the car is approaching

(b) the car is moving away.

Answer

(a) $$f_o = \frac{f_S v}{v - v_S}$$

$$f_o = \frac{1200\ \text{Hz} \times 340\ \text{m s}^{-1}}{340\text{m s}^{-1} - 35\ \text{m s}^{-1}} = 1300\ \text{Hz (2 sf)}$$

(b) $$f_o = \frac{f_S v}{v + v_S}$$

$$f_o = \frac{1200\ \text{Hz} \times 340\ \text{m s}^{-1}}{340\text{m s}^{-1} + 35\ \text{m s}^{-1}} = 1100\ \text{Hz (2 sf)}$$

22. A motorcycle is moving at a constant speed of 28 m s⁻¹ when the rider sounds a horn with a frequency of 1300 Hz.

Calculate the frequency of the horn perceived by an observer when:

(a) the motorcycle is approaching the observer

(b) the motorcycle is moving away from the observer.

The speed of sound in air is 340 m s⁻¹.

★**23.** The sound produced by the engine of a train has a wavelength of 17.4 cm when emitted in air. An observer sees the train approaching at a speed of 25.0 ms⁻¹.

(a) Calculate the wavelength of this sound as received by the observer.

(b) Calculate the frequency of this sound as received by the observer.

Take the speed of sound in air at the temperature in this location as 351 m s⁻¹.

Worked example

The driver of a train, which is moving at constant velocity, sees a person crossing the track. The driver sounds a horn that operates at 1350 Hz. The person close to the track hears the horn at a frequency of 1450 Hz. The speed of sound in air is 343 ms⁻¹.

(a) Calculate the speed of the train.

(b) Describe qualitatively how another person standing 50 m away from the track will hear the horn when the train is getting closer, compared to the driver and the person crossing the track.

Answer

(a) $f_o = \dfrac{f_S v}{(v - v_S)}$

$f_o (v - v_S) = f_S v$

$f_o v - f_o v_S = f_S v$

$f_o v_S = f_o v - f_S v$

$v_S = \dfrac{(f_o v - f_S v)}{f_o}$

$$v_S = \frac{1450\ \text{Hz} \times 343\ \text{m s}^{-1} - 1350\ \text{Hz} \times 343\ \text{m s}^{-1}}{1450\ \text{Hz}}$$

$= 23.7\ \text{m s}^{-1}$

(b) As the other person is 50 m from the track, the train will not be coming straight towards them, so they will observe a frequency higher than heard by the driver but not as high as the person directly in front of the train.

24. A motorcycle horn has a frequency of 1.1 kHz. An observer sees the motorcycle approaching and hears the horn with an apparent frequency of 1.2 kHz.
Calculate the speed of the motorcycle.
The speed of sound in air is 340 m s^{-1}.

25. An observer standing 15 m from a road sees a car approaching that is 64 m away travelling at 39 m s^{-1} (Figure 7.31).

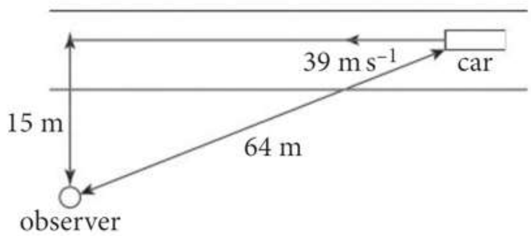


Figure 7.31

The car sounds a horn of frequency 1.3 kHz. Calculate the frequency of the horn as heard by the observer.

Key ideas

- If a source of sound waves is moving with a speed that is a significant proportion of the wave speed, this will cause changes in the properties of the wave observed.
- When the source of sound is moving towards a stationary observer, the observer will detect an increase in frequency of the sound compared with that emitted by the source.
- When the source of sound is moving away from a stationary observer, the observer will detect a decrease in frequency of the sound compared with that emitted by the source.
- An increase in frequency is heard as a higher pitch.
- A decrease in frequency is heard as a lower pitch.

Assignment 7.1: The Doppler effect in medicine

Background

Modern medicine involves more techniques that are non-invasive than in the past. Non-invasive means that no breaks in the skin are required, so the risk of infection and patient injury are greatly reduced compared with invasive techniques. Ultrasound is one of the resources available for non-invasive medical techniques and has many applications. The use of ultrasound in medicine is called sonography.

Ultrasound

Ultrasound is defined as sound waves that have higher frequencies than the upper limit of human hearing, which is 20 kHz.

A1. Calculate the wavelength of ultrasound that has a frequency of 24 MHz in air. The speed of sound in air is 340 m s^{-1}.
When used to study conditions inside the body, ultrasound has to travel through soft tissues. Soft tissues include those in skin, muscle and fat. The average speed of ultrasound waves in soft tissues is 1500 m s^{-1}.

A2. **(a)** Explain why the speed of ultrasound waves is higher in soft tissues than in air.
(b) Calculate the wavelength of ultrasound frequency 24 MHz in soft tissue.

Ultrasound and blood flow

Ultrasound can be used to determine the speed of blood flow in arteries and veins. Blood in a vein flows at an approximately constant speed. Blood in an artery flows at a speed that varies with the pulse rate in the artery. The pulse rate in an artery is the same as the rate of heart beats.

A3. A typical resting adult heart rate is 73 beats per minute. Calculate the frequency in Hz of the change in blood flow speed as a result of this heart rate.

Figure 7.32 shows how ultrasound can be used to measure the speed of blood flow.

The probe in Figure 7.32 transmits ultrasound at a frequency of f_S and receives reflected ultrasound at a frequency of f_o.

A4. Suggest why the ultrasound probe in Figure 7.32 must be in contact with the surface of the skin when in use.

The ultrasound is reflected from the red blood cells in the blood vessel.

A5. Explain, with reference to Figure 7.32, whether f' will be greater or less than f_o

The formula for the change in frequency, Δf, as a result of blood flow in Figure 7.32 is

$$\Delta f = \frac{2 f_o \, v_{rbc} \cos\theta}{v}$$

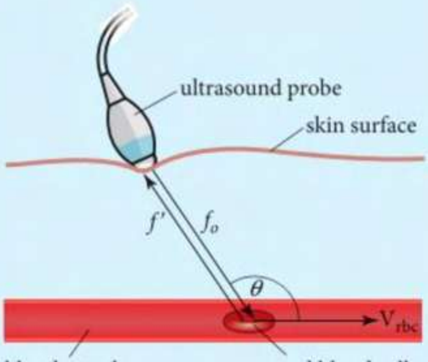


Figure 7.32 Use of ultrasound to measure the speed of blood flow.

where v_{rbc} is the speed of the red blood cells, θ is the angle between the direction of the transmitted waves and the direction of blood flow and v is the speed of ultrasound in the tissue.

A6. Suggest why f_0 is multiplied by 2 in the equation.

A7. There will also be reflections of the ultrasound waves from boundaries between tissues like the walls of the blood vessel. Explain why these do not affect the formula given above.

A8. A probe is used at an angle $\theta = 120°$ to blood flow in a blood vessel, with a transmitted wave frequency of 24 MHz. Calculate the reflected wave frequency change when the red blood cells are moving away from the probe at 10.5 cm s^{-1}. The speed of ultrasound in soft tissue is 1500 m s^{-1}.

In practice, blood flow in veins is turbulent. Turbulent flow is when the red blood cells and other contents are not travelling parallel to the walls of the blood vessel or even in straight lines. They can be travelling in curved paths or even, for short times, travelling in the opposite direction.

A9. Suggest how the measurement technique can be designed and the results analysed to take account of this turbulent flow.

The use of the Doppler effect with ultrasound to measure the speed of blood flow in veins was developed in the 1960s. In the 1990s, the technique was further developed to study blood flow in the brain. This technique is called transcranial Doppler (TCD).

Sound waves, and ultrasound, travel faster in more dense solids than in less dense ones. However, more dense materials greatly lower the intensity of the sound waves. There are also many reflections generated from tissue boundaries. Hence, TCD does not give accurate results for the speed of blood flow in blood vessels in the brain.

A10. Suggest the greater difficulties with using ultrasound to measure the speed of blood flow deep inside the brain compared to close to the skin surface that make TCD inaccurate for this purpose.

7.4 The electromagnetic spectrum

Link

Look back to topic 7.1 for features of transverse progressive waves.

The word **spectrum** is used to describe a range of wavelengths (or frequencies) of any one type of wave. The electromagnetic spectrum is a range of **electromagnetic waves**.

Electromagnetic waves are different to sound waves in that they are transverse. They have another important difference, in that they do not require a medium in which to travel. So they will travel in a vacuum, which is a space containing no matter such as atoms or molecules.

Progressive electromagnetic waves transfer energy from a source (where they are emitted) to a receiver (where they are absorbed).

Since electromagnetic waves will travel in a vacuum, this implies that there are no particles oscillating in the wave. Instead, an **electric field** and a **magnetic field** oscillate perpendicular to each other, with the same frequency. This is shown in Figure 7.33.

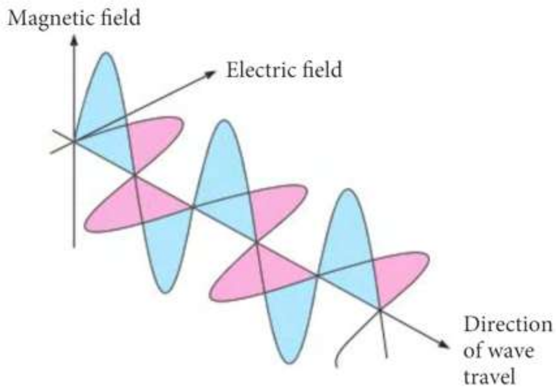


Figure 7.33 An electromagnetic wave is a transverse wave where an electric field (shown in pink) and a magnetic field (shown in blue) oscillate perpendicular to each other and perpendicular to the direction of wave travel.

Electromagnetic waves have a vast range of wavelengths, from more than 10 km to less than 10 pm. The electromagnetic spectrum is classified into seven main parts according to how these different wavelengths interact with matter when absorbed. The electromagnetic spectrum is shown in Figure 7.34.

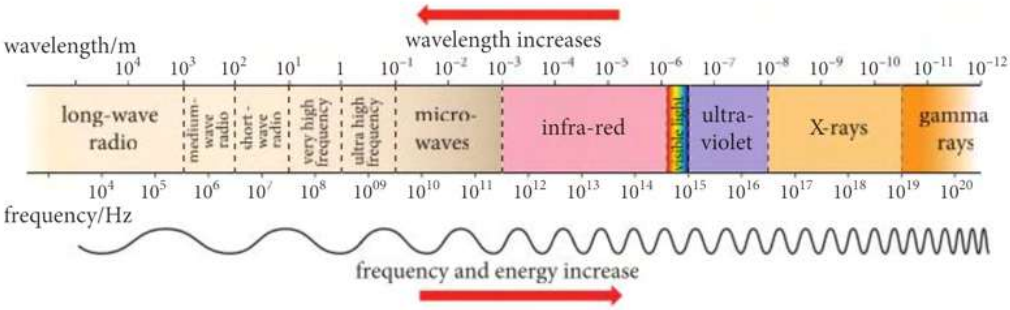


Figure 7.34 The electromagnetic spectrum.

The wavelengths of the main regions of the electromagnetic spectrum are:

- Radio waves 10^{-1} – 10^{4} m
- Microwaves 10^{-3} – 10^{-1} m.
- Infra red waves 700 nm–10^{-3} m
- Visible light waves 400 – 700 nm
- Ultra violet waves 10^{-8} m – 400 nm
- X-rays 5×10^{-11} m – 10^{-8} m
- Gamma rays (γ rays) 10^{-12} – 5×10^{-11} m

(Note: The X-ray and gamma ray ranges overlap – the nature of the radiation depends upon the origin of the rays). Visible light waves are so-called because they are detected by the human eye. X-rays and γ-rays are so-called because it was not known that they were waves when they were discovered.

All electromagnetic waves are transverse and travel at the same speed in a vacuum. This speed is predicted by the theory of special relativity to be the maximum speed of any particle in the Universe. The theory predicts that only particles with zero mass can travel at this speed and this speed is impossible for other objects to reach. The speed is often called the speed of light. Its value is 3.00×10^{8} m s^{-1}, which is 1080 million km h^{-1}, and it is denoted by *c*.

The speed of electromagnetic waves in air has this same value (to this degree of precision). The speed in other mediums is lower.

Electromagnetic waves, as all waves, obey the wave equation $v = f\lambda$. Therefore, for electromagnetic waves in free space,

$$c = f\lambda$$

Tip

There is no physical difference between the named types of electromagnetic waves, other than their frequency and wavelength.

Link

You can learn in Topic 22.1 that electromagnetic energy is delivered in packets called photons whose energy is proportional to the wave frequency.

Link

See Table 7.1 in Topic 7.1 for a comparison of wave speeds.

The range in wavelengths of the different types of electromagnetic waves can be expressed as a range in frequencies, as shown in the spectrum in Figure 7.34. Waves with a large wavelength, such as radio waves, have low frequencies. Waves with small wavelengths, such as **gamma rays**, have high frequencies. The wave frequency is the frequency of oscillation of the electric and magnetic fields in the electromagnetic wave.

At A Level you will learn that electromagnetic waves can also show particle properties, and the particles that carry electromagnetic energy are called photons. The higher the wave frequency, the greater the energy carried. The reason that photons can travel at the speed of light is that they have zero mass.

Tip

The term free space is sometimes used to mean a vacuum.

Link

You will learn more about the particle properties of electromagnetic waves in Chapter 22 and also discover the experimental evidence for these particle properties.

Key ideas

- Electromagnetic waves are transverse.
- Electromagnetic waves have an electric field and a magnetic field oscillating perpendicular to each other and perpendicular to the direction of travel of the wave.
- The frequencies of oscillation of the electric and magnetic fields are equal and represent the frequency of the wave.
- All electromagnetic waves travel at the same speed, c, in a vacuum.
- The wave equation for electromagnetic waves is $c = f\lambda$.
- Wavelengths of electromagnetic waves range from more than 10^4 m (radio waves) to 10^{-12} m (gamma rays).

26. The electromagnetic spectrum covers wavelengths from 10 km to 10 pm. Calculate how many times longer 10 km is than 10 pm.

27. State the range of wavelengths from the electromagnetic spectrum in air that is visible to the human eye.

28. Name the part of the electromagnetic spectrum where waves of these wavelengths would be found:
(a) 1.4 km
(b) 650 nm
(c) 3.7 cm

29. Calculate the frequency of each of these wavelengths in the electromagnetic spectrum. Give your answers using suitable unit prefixes and not in standard form.
(a) 640 m
(b) 10 cm
(c) 1.0 μm

30. What part of the electromagnetic spectrum has waves of wavelength approximately equal to
(a) the wavelength of the sound of the human voice when speaking
(b) the diameter of a carbon atom
(c) the width of a fine pencil line?

7.5 Polarisation

Polarisation is a phenomenon associated only with transverse waves. A transverse wave has oscillations perpendicular to the direction of travel of the wave. All types of transverse waves can show polarisation. Longitudinal waves cannot show polarisation.

Observation of polarisation of light waves has given evidence that electromagnetic waves are transverse. Figure 7.33 in section 7.4 shows a single electromagnetic wave in which the electric field is oscillating only in the horizontal plane. Such a wave is described as being **plane polarised**, in this case horizontally polarised. In electromagnetic waves, the plane of polarisation always refers to the plane in which the electric field oscillates.

Some light sources, such as some lasers and liquid crystal displays (LCD) emit polarised light. Light from a lamp, or from the Sun, is unpolarised. Unpolarised means that the electric fields of all the waves are oscillating randomly, in all possible planes perpendicular to the wave direction. This is shown in Figure 7.35.

Link

In Chapter 8 you will see evidence for the wave nature of light. In Chapter 22 we will also explore the evidence for the particulate nature of electromagnetic radiation.

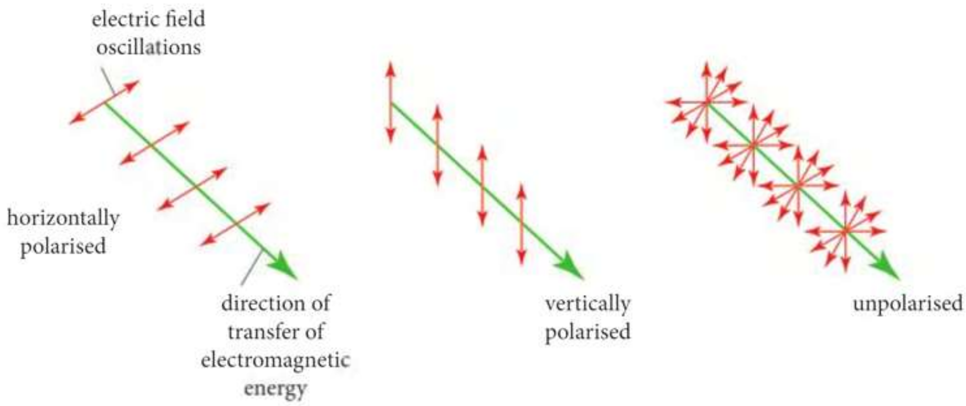


Figure 7.35 Unpolarised electromagnetic radiation is composed of waves with their electric fields oscillating in random directions perpendicular to the direction of travel.

In Figure 7.35, the double-headed arrows show the directions of oscillation of the electric fields.

POLARISING LIGHT

Polarised light can be produced from unpolarised light by passing through a **polarising filter**. A polarising filter is made of a material such as Polaroid that only allows light polarised in one particular plane to pass through. The direction of polarisation allowed by the filter is called the axis of polarisation of the filter.

Incident light that has no component of its electric field strength along the axis of polarisation will be absorbed by the filter. The rest of the incident light passing through the filter emerges polarised, with its plane of polarisation corresponding to the direction of the filter's axis of polarisation. The intensity of the polarised light transmitted through the polarising filter will be approximately half the intensity of the incident unpolarised light.

For example, in Figure 7.36 both filters A have their axis of polarisation vertical. This is modelled by showing the wave passing through a vertical slit. The transmitted wave has its plane of polarisation vertical.

The use of two successive polarising filters allows the intensity of the transmitted beam to be reduced by a varying degree.

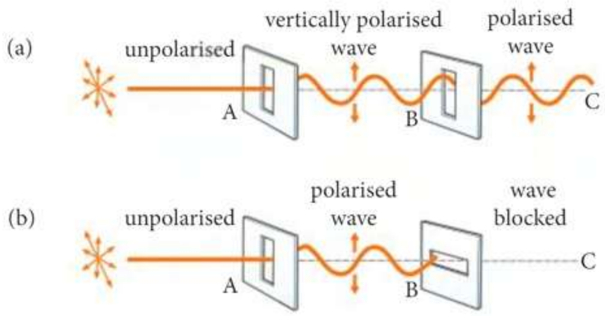


Figure 7.36 (a) Filters A and B both have vertical axes of polarisation and so light arriving at C is vertically polarised. (b) Filter A allows vertically polarised light through, but filter B has a horizontal axis of polarisation, so no light arrives at C.

In Figure 7.36 (a) the axis of polarisation of filter B is in the same direction as that of the plane of polarisation of the light incident on it. In other words, the angle between the axis of polarisation of filter B and the plane of polarisation of the light is zero. In this case, filter B does not absorb any of the light. The light intensity transmitted by the filter is the same as that incident on it.

In Figure 7.36 (b) the angle between the axis of polarisation of filter B and the plane of polarisation of the light is 90°. In this case, filter B absorbs all of the light, and the intensity arriving at C is zero.

At values of the angle between zero and 90°, some light will be transmitted though the filter. The plane of polarisation of the transmitted light will match the direction of the axis of polarisation of the second filter (Figure 7.37).

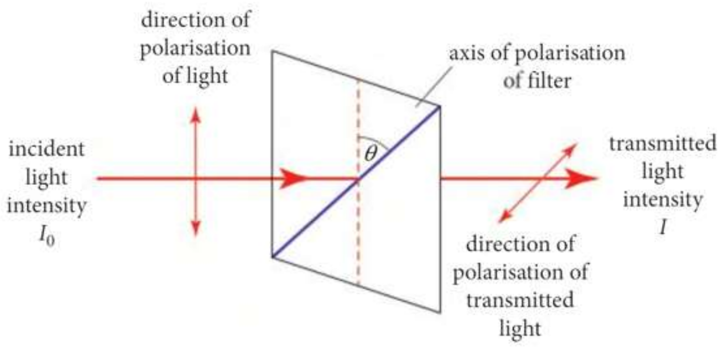


Figure 7.37 The polarised light is reduced in intensity and its plane of polarising has been rotated

Tip

$\cos^2\theta = (\cos\theta)^2$ so in calculations find the cosine of the angle in degrees, then square the result.

The intensity, I of light transmitted through a polarising filter is related to the initial intensity, I_0 of polarised light incident on the filter and the angle θ between the polarisation of the light and the axis of polarisation of the filter by **Malus' law**:

$$I = I_0 \cos^2\theta$$

Worked example

Light of intensity 250 W m^{-2} is horizontally polarised. The light is incident on a filter with an axis of polarisation inclined at 35° to horizontal.
Calculate the intensity of the light transmitted by the filter.

Answer

$I = I_0 \cos^2\theta$
$= 250\ \text{W m}^{-2} \times (\cos 35°)^2 = 168\ \text{W m}^{-2}$

Link

Look back to Topic 7.1 for a definition of the intensity of a wave.

31. Light that is polarised horizontally has an intensity of 1800 W m^{-2}. This light passes through a polarising filter that is inclined at 45° to the plane of polarisation of the light. Calculate the intensity of the light emerging from the filter.

32. Unpolarised light passes through a polarising filter, A. The light from filter A then passes through another polarising filter, B, with its axis of polarisation inclined at 30° to that of filter A. The intensity of the light emerging from filter B is 200 W m^{-2}. Calculate the intensity of the unpolarised light arriving at filter A.

Worked example

Light polarised in the vertical plane is incident on a filter with an axis of polarisation that is not vertical. The filter transmits 25% of intensity of the incident light.

Calculate the angle between the axis of polarisation of the filter and the vertical plane.

Answer

$$I = I_o \cos^2\theta$$

$$\cos^2\theta = \frac{I}{I_o}$$

$$\cos\theta = \sqrt{\frac{I}{I_o}}$$

$$\theta = \cos^{-1}\sqrt{\frac{I}{I_o}}$$

$$= \cos^{-1}\sqrt{\frac{1}{4}}$$

$$= \cos^{-1} 0.5$$

$$= 60°$$

33. A polarising filter reduces the intensity of polarised light by 25%. Calculate the angle between the axis of polarisation of the filter and the plane of polarisation of the light.

APPLICATIONS OF POLARISATION

Reducing glare

When unpolarised light, for example from the sun or from a lamp, is reflected by a non-metallic surface such as water or glass, the reflected light is partially plane polarised. Partially plane polarised means that a significant proportion of the light is polarised in one plane.

Partial polarisation occurs on reflection because incident light that has its electric field in a plane perpendicular to the surface is partially absorbed. The plane of polarisation of most of the reflected light is therefore parallel to the reflecting surface.

Polaroid sunglasses act as polarising filters with an axis of polarisation perpendicular to large reflecting surfaces such as the road and the sea. Much of the polarised light from such reflections is absorbed, thereby reducing glare on a sunny day.

LCD displays emit polarised light. Hold the lens of Polaroid sunglasses over the LCD display of a calculator so that you can see the display brightly. Rotate the lens through 90° and the display appears to go completely black. This can be explained by Malus' law, as you have changed the angle between the plane of polarisation and the axis of polarisation of the filter from 0 to 90°.

Figure 7.38 The photograph on the right was taken using a polarising filter.

Link

Interference between waves and the conditions needed to produce interference is covered in Chapter 8.

Photographers also use polarising filters to reduce glare (Figure 7.38), or to eliminate reflections for example when taking a photograph through a glass window.

Radio reception

Radio waves that are used to transmit TV signals are polarised for two reasons. It is more efficient for a transmitter to emit waves only polarised in one plane. Also, it can reduce **interference** from other TV transmitters if those are polarised in a different plane.

Receiving aerials for TV signals must be pointed towards the transmitter and must also be aligned in the same plane of polarisation as the signal (Figure 7.39).

Figure 7.39 This TV receiving aerial is aligned to receive horizontally polarised signals.

Drug testing

Some molecules, when dissolved in water, can rotate the plane of polarised light that is passed through the solution. These molecules each have two versions, called isomers, that will rotate the plane of polarised light in different directions. A mixture of the two isomers of the same compound will not rotate the plane of polarised light. This allows an important test in the manufacture of some pharmaceutical drugs. The active molecule in a drug may have one isomer that works well and another isomer that can give severe side effects.

Key ideas

- Polarisation is possible with all transverse waves.
- A transverse wave is plane polarised if the oscillations are only in one plane.
- In electromagnetic waves, the plane of oscillation of the electric field defines the plane of polarisation.
- Polarising filters can be used to polarise unpolarised light, or to reduce the intensity of polarised light.
- Malus' law can be used to calculate the intensity of light after polarised light passes through a polarising filter: $I = I_0 \cos^2\theta$
  where θ is the angle between the polarisation of the incident light and the axis of polarisation of the filter.
- The transmitted light is polarised in the same direction as the axis of polarisation of the filter.

34. Which of these types of wave cannot be polarised?

A light

B sound

C X-rays

D gamma rays

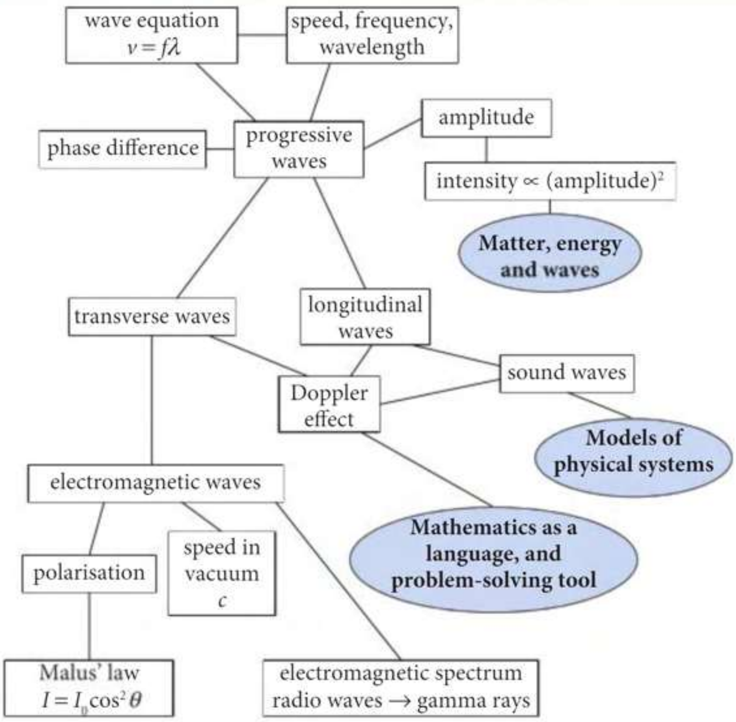
35. Give two reasons why Polaroid lenses do not filter out all reflections of light on a sunny day.

36. Light of intensity 850 W m^{-2} which is polarised in the horizontal plane is incident on a polarising filter, A. The axis of polarisation of filter A is inclined at 35° to horizontal.

(a) Calculate the intensity of the light transmitted from polarising filter A.

(b) Another filter, B, inclined at 45° to horizontal is inserted after filter A. Calculate the intensity of the light transmitted from filter B.

CHAPTER OVERVIEW



WHAT YOU HAVE LEARNED

- Describe what is meant by wave motion as illustrated by vibration in ropes, springs and ripple tanks.
- Understand and use the terms displacement, amplitude, phase difference, period, frequency, wavelength and speed in the context of wave motion.
- Understand the use of the time-base and y-gain of a cathode-ray oscilloscope (CRO) to determine frequency and amplitude.
- Derive the wave equation $v = f\lambda$ starting from the definitions of speed, frequency and wavelength.
- Recall and use the equation $v = f\lambda$ in calculations about waves.
- Understand that energy is transferred by a progressive wave.
- Remember and use the relationships: intensity = power ÷ area; and intensity ∝ (amplitude)^2 for a progressive wave.
- Compare the properties of transverse and longitudinal waves.

- Analyse and interpret graphs of transverse and longitudinal waves.
- Understand that when there is relative movement between a source of sound waves and an observer, there is a change in observed frequency and wavelength due to the Doppler effect.
- Use the expression $f_o = \frac{f_S v}{v \pm v_S}$ for the observed frequency when a source of sound waves moves relative to an observer.
- State that all electromagnetic waves are transverse waves that travel with the same speed c in free space.
- Recall the approximate range of wavelengths of the principal electromagnetic radiations from radio waves to γ-rays.
- Recall that wavelengths in the range 400–700 nm in free space are visible to the human eye.
- Understand that polarisation is a phenomenon associated with transverse waves.
- Recall and use Malus' law ($I = I_0 \cos^2\theta$) to calculate the intensity of a plane polarised electromagnetic wave after transmission through a polarising filter or a series of polarising filters.

CHAPTER REVIEW

1. Which of these waves transfer energy in their direction of travel?
   A only electromagnetic waves
   B all progressive waves
   C transverse waves only
   D longitudinal waves only
2. Which of these can travel in free space?
   A all transverse waves
   B all longitudinal waves
   C all electromagnetic waves
   D all sound waves
3. **(a)** Compare the difference in direction of oscillations in transverse and longitudinal waves.
   (b) State what oscillates in:
   (i) a sound wave in air
   (ii) an electromagnetic wave in a vacuum.
4. Give one named example of:
   (a) a longitudinal wave
   (b) an electromagnetic wave.
5. Name the part of the electromagnetic spectrum that contains waves with these wavelengths.
   (a) 1 nm
   (b) 10 μm
   (c) 650 nm
   (d) 2.5 km
   (e) 1.8 cm
6. A TV signal is vertically polarised. Explain what this means.
7. Name the device that is capable of:
   (a) producing a sine wave with constant amplitude and variable frequency
   (b) displaying a graphical representation of a wave on a screen
   (c) producing water waves for display on a screen
   (d) converting sound waves into electrical signals.

8. (a) State the wave equation.
   (b) Use the wave equation to complete Table 7.4. Use an appropriate number of significant figures.

type of wave	frequency / Hz	wavelength / m	speed / m s^{-1}
sound	1.2×10^4		340
water		5	9
ultrasound	9.5×10^4	1.6×10^{-2}	
electromagnetic		6.81×10^{-7}	3.00×10^8

Table 7.4

9. (a) Name the effect that causes an apparent shift in frequency when a source of sound waves is moving relative to an observer.
   (b) A motorcycle engine emits a sound of frequency 2.4 kHz. The speed of sound in air is 340 m s^{-1}. Calculate the apparent frequency of this sound to an observer when
      (i) the motorcycle is approaching at 28 m s^{-1}
      (ii) the motorcycle is moving away at 33 m s^{-1}.

10. (a) Polarised light is incident on a polarising filter. The axis of polarisation of the filter is inclined at 40° to the plane of polarisation of the light. Calculate the ratio

$$\frac{\text{intensity of light transmitted by the filter}}{\text{intensity of light incident on the filter}}$$

   (b) A student wants to rotate the plane of polarised light by 90°.
   The student has a choice of two methods.
   Method A. Using three successive filters, each rotating the plane of polarisation by 30°.
   Method B. Using two successive filters, each rotating the plane of polarisation by 45°.
   Show that method A will give a higher intensity of light than method B.